

HW 10

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PHYS116
Homework 10

Townsend: 10.3, 10.7, 10.8, 10.19, 11.1, 11.4

Townsend Problem 10.3 (5 pts)

A negatively charged pion (a spin-0 particle) is bound to a proton forming a pionic hydrogen atom. At time $t = 0$ the system is in the state

$$|\psi\rangle = \frac{1}{2}|1,0,0\rangle + \frac{1}{\sqrt{2}}|2,1,1\rangle + \frac{1}{2}|2,1,0\rangle$$

In an external magnetic field in the z direction, the Hamiltonian is given by

$$\hat{H} = \frac{\hat{\mathbf{p}}^2}{2\mu} - \frac{e^2}{|\hat{\mathbf{r}}|} + \omega_0 \hat{L}_z$$

where μ is the reduced mass of the pion-proton system.

(a) What is $|\psi(t)\rangle$ for the system at time t ? What is $\langle E \rangle$ at time t for this state?

(b) What is $\langle L_z \rangle$ and $\langle L_x \rangle$ at time t for this state?

A

We have the E for $|E, L, M\rangle$ given to us, we just time evolve each portion.

$$|\psi(t)\rangle = \frac{1}{2}e^{-\frac{i}{\hbar}E_1 t}|1,0,0\rangle + \frac{1}{\sqrt{2}}e^{-\frac{i}{\hbar}E_2 t}|2,1,1\rangle + \frac{1}{2}e^{-\frac{i}{\hbar}E_2 t}|2,1,0\rangle$$

The Hamiltonian looks like

$$\hat{H} = \hat{H}_{\text{hydrogen}} + \omega_0 \hat{L}_z$$

The Hamiltonian of the Hamilton has states $|n, l, m\rangle$, which is also an eigenstate of $\hat{L}_z$.

This gives us the time evolution as

$$e^{-i\hat{H}_{\text{hydrogen}} t - \omega_0 \hat{L}_z t}$$

E_n for hydrogen is

$$\frac{\mu z^2 e^4}{2\hbar^2 n^2}$$

where

$$E_1 = \frac{-\mu e^4}{2\hbar^2}$$

$$E_2 = \frac{-\mu e^4}{8\hbar^2}$$

The eigenvalues for these are

$$|1,0,0\rangle \text{ has } E = E_1$$

$$|2,1,1\rangle \text{ has } E = E_2 - \omega_0 \hbar$$

$$|2,1,0\rangle \text{ has } E = E_2$$

So the time evolution is

$$\psi(t) = \frac{1}{2}e^{-i\frac{E_1}{\hbar}t}|1,0,0\rangle + \frac{1}{\sqrt{2}}e^{-i(\frac{E_2}{\hbar} + \omega_0)\hbar t}|2,1,1\rangle + \frac{1}{2}e^{-i\frac{E_2}{\hbar}t}|2,1,0\rangle$$

$$\begin{aligned}
\langle E \rangle &= \langle \psi | H | \psi \rangle = \frac{1}{4} \langle E_1 \rangle + \frac{1}{2} \langle E_2 + \omega_0 \hbar \rangle + \frac{1}{4} \langle E_2 \rangle \\
&= -\frac{\mu e^4}{8 \hbar^2} - \frac{\mu e^4}{16 \hbar^2} + \frac{\omega_0 \hbar}{2} - \frac{\mu e^4}{32 \hbar^2} \\
&= -\frac{\mu e^4}{\hbar^2} \left(\frac{1}{8} - \frac{1}{16} - \frac{1}{32} \right) + \frac{\omega_0 \hbar}{2} \\
&= -\frac{\mu e^4}{\hbar^2} \left(\frac{4}{32} - \frac{2}{32} - \frac{1}{32} \right) + \frac{\omega_0 \hbar}{2} \\
&= \frac{\omega_0 \hbar}{2} - \frac{1}{32} \frac{\mu e^4}{\hbar^2}
\end{aligned}$$

B

TODO

Townsend Problem 10.7 (5 pts)

An electron is in the ground state of tritium, for which the nucleus is the isotope of hydrogen with one proton and two neutrons. A nuclear reaction instantaneously changes the nucleus into ^3He , which consists of two protons and one neutron. Calculate the probability that the electron remains in the ground state of the new atom. Obtain a numerical answer.

Note: It is sufficient to treat the mass of the proton and neutron as equal.

We know that the Hamiltonian for hydrogen is

$$\hat{H} = \frac{P^2}{2m} + \frac{ze^2}{|r|}$$

For tritium, we have a different way of making the same mass, but the mass is constant because the proton and neutron are (for all intents) equal masses. We just have a different charge -- z goes from $+1$ to $z = +2$.

$$a_0 = \frac{\hbar}{\mu c \alpha} \quad (10.41)$$

$$R_{1,0} = 2 \left(\frac{Z}{a_0} \right)^{3/2} e^{-Zr/a_0} \quad (10.43)$$

For the tritium, this is

$$R_{1,0} = 2 \left(\frac{1}{a_0} \right)^{3/2} e^{-\frac{1r}{a_0}}$$

and for the Helium isotope, we have

$$R_{1,0} = 2 \left(\frac{2}{a_0} \right)^{3/2} e^{-\frac{2r}{a_0}}$$

Assuming separation of variables, these also have Y_{lm} angular (θ and ϕ) dependence which is the exact same, $Y_{0,0} = \sqrt{\frac{1}{4\pi}}$.

We integrate these over all space to find the inner product (each of these is $\langle \psi | r \rangle$, and real so $\langle r | \psi \rangle$ too) (theta is around the xy plane, ϕ is translating up and down)

$$\begin{aligned}
&\frac{1}{\pi} \cdot \left(\frac{2}{a_0^2} \right)^{3/2} \int_{r=0}^{\infty} \int_{\phi=0}^{2\pi} \int_{\theta=-\frac{\pi}{2}}^{\frac{\pi}{2}} dr d\theta d\phi \, r^2 \sin \theta \, e^{-\frac{2r}{a_0}} e^{-\frac{r}{a_0}} \\
&= -4 \left(\frac{2}{a_0^2} \right)^{3/2} \int_{r=0}^{\infty} r^2 e^{-\frac{3r}{a_0}} dr \\
&= -\frac{16}{27} \sqrt{2} \left(\frac{1}{a_0^2} \right)^{3/2} a^3
\end{aligned}$$

This is the amplitude. The probability of finding the ground state after the reaction is

$$= \left(-\frac{16}{27} \sqrt{2} \left(\frac{1}{a_0^2} \right)^{\frac{3}{2}} a^3 \right)^2$$

$$= 0.702$$

Townsend Problem 10.8 (5 pts)

Show that there are no allowed energies $E < -V_0$ for the potential well

$$V = \begin{cases} -V_0 & r < a \\ 0 & r > a \end{cases}$$

by explicitly solving the Schrödinger equation and attempting to satisfy all the appropriate boundary conditions.

Note: $\ell = 0$ corresponds to the ground state. Start by solving the radial equation for $u(r) = rR(r)$ with $\ell = 0$ for $r < a$ and $r > a$ and then, as the problem indicates, attempting to satisfy the appropriate boundary conditions, namely at $r = 0$, $r = a$, and r approaching infinity.

at $r = 0$, $r = 1$ we should get ???

at $r \rightarrow \infty$ we should get probability amplitude 0.

the radial equation is

$$\left[-\frac{\hbar^2}{2\mu} \frac{d^2}{dr^2} + \frac{l(l+1)\hbar^2}{2\mu r^2} + V(r) \right] u(r) = E u(r) \quad (9.98)$$

We have

$$\left[-\frac{\hbar^2}{2\mu} \frac{d^2}{dr^2} + \frac{l(l+1)\hbar^2}{2\mu r^2} + V(r) \right] rR(r) = E rR(r)$$

with $L = 0$, we can look inside the well and outside it

case 1: in the well, r less than a

$$\left[-\frac{\hbar^2}{2\mu} \frac{d^2}{dr^2} - v_0 \right] rR(r) = E rR(r)$$

TODO

for r greater than a

$$\left[-\frac{\hbar^2}{2\mu} \frac{d^2}{dr^2} \right] rR(r) = E rR(r)$$

This has solutions with $k = \sqrt{\frac{\hbar^2}{2\mu}}$

$k \sin(r)$ and $k \cos(r)$

Townsend Problem 10.19 (5 pts)

The Hamiltonian for two spin- $\frac{1}{2}$ particles, one with mass m_1 and the other with mass m_2 , is given by

$$\hat{H} = \frac{\hat{\mathbf{p}}_1^2}{2m_1} + \frac{\hat{\mathbf{p}}_2^2}{2m_2} + V_a(|\hat{\mathbf{r}}|) + \left(\frac{1}{4} - \frac{\hat{\mathbf{S}}_1 \cdot \hat{\mathbf{S}}_2}{\hbar^2} \right) V_b(|\hat{\mathbf{r}}|)$$

where $\hat{\mathbf{r}} = \hat{\mathbf{r}}_1 - \hat{\mathbf{r}}_2$ and

$$V_a(r) = \begin{cases} 0 & r < a \\ V_0 & r > a \end{cases} \quad V_b(r) = \begin{cases} 0 & r < b \\ V_0 & r > b \end{cases}$$

with $b < a$ and V_0 very large and positive.

(a) Determine the normalized position-space energy eigenfunction for the ground state. What is the spin state of the ground state? What is the degeneracy? Note: Take V_0 to be infinite.

(b) What can you say about the energy and spin state of the first-excited state? Does your result depend on how much larger a is than b ? Explain.

Note: What are the eigen states of the spin-dependent term in parenthesis? Each of these spin configurations can have its own position-space wave function associated with it.

TODO

For the problem with the two spherical potentials, one labelled $V\{a\}$ and one labelled $V\{b\}$ I would strongly recommend drawing out your potential as a function of position under two different cases. You need to figure out what the cases are yourself, but I can tell you it has to do with the spin of your particles.

INTEGRATE TO EVALUATE THE EXPECTATION VALUE IN POSITION SPACE

$$E_n^{(1)} = -q \langle \mathbf{E} | \langle n | \hat{x} | n \rangle = -q \langle \mathbf{E} | \int_{-\infty}^{\infty} dx |\langle x | n \rangle|^2 x = 0$$

Townsend Problem 11.1 (5 pts)

Consider a perturbation $\hat{H}_1 = b\hat{x}^4$ to the simple harmonic oscillator Hamiltonian

$$\hat{H}_0 = \frac{\hat{p}_x^2}{2m} + \frac{1}{2}m\omega^2\hat{x}^2$$

This is an example of an anharmonic oscillator, one with a nonlinear restoring force.

(a) Show that the first-order shift in the energy is given by

$$E_n^{(1)} = \frac{3\hbar^2 b}{4m^2\omega^2} (1 + 2n + 2n^2)$$

(b) Argue that no matter how small b is, the perturbation expansion will break down for some sufficiently large n . What is the physical reason?

a

$$E_n^{(1)} = \langle \varphi_n^{(0)} | \hat{H}_1 | \varphi_n^{(0)} \rangle \quad (11.15)$$

The $\psi^{(0)}$ is the wave function of the unperturbed SHO.

We have

$$\begin{aligned} & \langle N | \hat{H}_1 | N \rangle \\ &= \langle N | b\hat{x}^4 | N \rangle \end{aligned}$$

We can write $\hat{x}$ in terms of the raising and lowering operators

$$\hat{x} = \sqrt{\frac{\hbar}{2m\omega}} (\hat{a} + \hat{a}^\dagger) \quad (7.125)$$

This gives us

$$\left(\frac{\hbar}{2m\omega} \right)^2 b \langle N | (\hat{a} + \hat{a}^\dagger)^4 | N \rangle$$

We keep terms with equal raising and lowering,

$$\hat{a}^2\hat{a}^{2\dagger} + \hat{a}\hat{a}^\dagger\hat{a}\hat{a}^\dagger + \hat{a}\hat{a}^{2\dagger}\hat{a} + \hat{a}^{2\dagger}\hat{a}^2 + \hat{a}^\dagger\hat{a}\hat{a}^\dagger\hat{a} + \hat{a}^\dagger\hat{a}^2\hat{a}^\dagger$$

$$\hat{a}|n\rangle = \sqrt{n}|n-1\rangle$$

$$\hat{a}^\dagger|n\rangle = \sqrt{n+1}|n+1\rangle$$

Lets do each term ONE BY FRIGGEN ONE.

FORK.

1.

$$\begin{aligned} \hat{a}^2\hat{a}^{2\dagger}|n\rangle &= \hat{a}^2\sqrt{n+1}\sqrt{n+2}|n+2\rangle \\ &= (n+1)(n+2)|n\rangle \\ n^2 + 3n + 2 \end{aligned}$$

2.

$$\begin{aligned} \hat{a}\hat{a}^\dagger\hat{a}\hat{a}^\dagger|n\rangle &= (n+1)^2|n\rangle \\ n^2 + 2n + 1 \end{aligned}$$

3.

$$\begin{aligned} \hat{a}\hat{a}^{2\dagger}\hat{a}|n\rangle &= n(n+1)|n\rangle \\ n^2 + n \end{aligned}$$

4.
$$\hat{a}^{2\dagger}\hat{a}^2|n\rangle = \frac{n(n-1)}{n^2-n}|n\rangle$$

5.
$$\hat{a}^\dagger\hat{a}\hat{a}^\dagger\hat{a}|n\rangle = n^2|n\rangle$$

6.
$$\hat{a}^\dagger\hat{a}^2\hat{a}^\dagger|n\rangle = \frac{n(n+1)}{n^2+n}|n\rangle$$

$$\left(\frac{\hbar}{2m\omega}\right)^2 b(n^2 + 3n + 2 + n^2 + 2n + 1 + n^2 + n + n^2 - n + n^2 + n^2 + n)$$

which gives us

$$\left(\frac{\hbar}{2m\omega}\right)^2 b(6n^2 + 6n + 3)$$

b

Perturbation theory requires that the difference is small - if we have really large n , then this can be on the order of the energy without the perturbation. We wouldn't get a fine structure, but some wildly different thing. Its like Taylor expanding and relying on it far from the point of expansion.

Townsend Problem 11.4 (5 pts)

Calculate the first-order shift to the energy of the ground state and first-excited state of a particle of mass m in the one-dimensional infinite square well

$$V(x) = \begin{cases} 0 & 0 < x < L \\ \infty & \text{elsewhere} \end{cases}$$

of (a) the constant perturbation $\hat{H}_1 = V_1$ and (b) the linearly increasing perturbation $\hat{H}_1 = \varepsilon E_1^{(0)} x/L$, where $E_1^{(0)}$ is the unperturbed energy of the ground state and $\varepsilon \ll 1$.

The energy in the ground state of the square well is

$$\frac{n^2\pi^2\hbar^2}{2mL^2}$$

With the state $\psi_n(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right)$

a

We have a time independent perturbation

$$\langle \varphi_k^{(0)} | \hat{H}_0 | \varphi_n^{(1)} \rangle + \langle \varphi_k^{(0)} | \hat{H}_1 | \varphi_n^{(0)} \rangle = E_n^{(0)} \langle \varphi_k^{(0)} | \varphi_n^{(1)} \rangle$$

We know that the first order shift is just the expectation value, so we should end up with just V_1 . However, I want to do the process to make sure (I'm so shaky at quantum, I kind of lowkey hate this class).

Because we have the position space, we have to integrate over all space to get the inner product

$$\begin{aligned} & \frac{2}{L} \int_0^L \sin\left(\frac{\pi x}{L}\right) V_1 \sin\left(\frac{\pi x}{L}\right) dx \\ &= \frac{2V_1}{L} \int_0^L \sin^2\left(\frac{\pi x}{L}\right) dx \end{aligned}$$

Looking up the integral in wolfram because I know I could do it with a 1- cos sub and do stuff, and I'm lazy and already guessed the answer correctly (the integral is $\frac{L}{2}$ which cancels with $\frac{2}{L}$)

$$= V_1$$

So the perturbed energies are

$$\frac{n^2 \pi^2 \hbar^2}{2mL^2} + V_1$$

TODO find the state

B

$$\hat{H}_1 = \epsilon E_0 \frac{\hat{x}}{L}$$

We can find the expectation value of this as

$$\frac{2}{L} \frac{1}{L} \frac{\pi^2 \hbar^2}{2mL^2} \int_0^L \sin^2 \left(\frac{\pi x}{L} \right) x dx$$

I'm running low on time again (why do I keep doing these the morning of RAWR) so I plugged into wolfram - it goes to $\frac{L^2}{4}$

The new perturbed energy in the ground state is

$$E_0 = \frac{1}{4} \frac{\pi^2 \hbar^2}{mL^2}$$

TODO find the state